Additional Topics

1 Time Series Regressions with Exogenous Regressors

1.1 Finite Sample Properties of OLS

- Consider the model

$$y_t = \beta_0 + \beta_1 x_{1t} + \ldots + \beta_k x_{kt} + u_t, \quad \text{for } t = 1, 2, \ldots, T.$$

  Let $\mathbf{x}_t = (x_{1t}, x_{2t}, \ldots, x_{kt})'$, and $\mathbf{X}$ denotes the collection of all independent variables across all time periods.

- **Gauss-Markov Theorem**: Under Assumptions 1–5, the OLS estimators are the *best linear unbiased estimators* (Wooldridge, 2019, Theorem 10.4).
  - Assumption 1. **Linear in parameters**
  - Assumption 2. **No perfect collinearity**
  - Assumption 3. **Zero conditional mean**:

$$E(u_t|\mathbf{X}) = 0, \quad \text{for } t = 1, 2, \ldots, T.$$

  - Assumption 4. **Homoskedasticity**:

$$Var(u_t|\mathbf{X}) = Var(u_t) = \sigma^2, \quad \text{for } t = 1, 2, \ldots, T.$$

  - Assumption 5. **No serial correlation**:

$$Corr(u_t, u_s|\mathbf{X}) = 0, \quad \text{for all } t \neq s.$$

- Assumption 3 is critical and requires more than contemporaneous exogeneity. When Assumption TS.3 holds, the explanatory variables are said to be *strictly exogenous*.
- The OLS estimators are unbiased under Assumptions 1–3.

1.2 Asymptotic Properties of OLS

- Under Assumptions 1'–3', the OLS estimators are consistent. Under Assumptions 1'–5', the OLS estimators are asymptotically normally distributed, and the usual OLS standard errors, t/F statistics are asymptotically valid. (Wooldridge, 2019, Theorems 11.1 and 11.2)
  - Assumption 1'. **Linearity and weak dependence**: Assumption 1 + the assumption that $\{y_t, \mathbf{x}_t\}_{t=1}^{T}$ is stationary and weakly dependent.
  - Assumption 2'. **No perfect collinearity**. Same as Assumption 2.
  - Assumption 3'. **Zero conditional mean**: $\mathbf{x}_t$ are *contemporaneously exogenous*,

$$E(u_t|\mathbf{x}_t) = 0, \quad \text{for } t = 1, 2, \ldots, T.$$

  - Assumption 4'. **Homoskedasticity**: The errors are *contemporaneously homoskedastic*:

$$Var(u_t|\mathbf{x}_t) = \sigma^2, \quad \text{for } t = 1, 2, \ldots, T.$$

  - Assumption 5'. **No serial correlation**:

$$E(u_t u_s|\mathbf{x}_t, \mathbf{x}_s) = 0, \quad \text{for all } t \neq s.$$

- The consistency result holds if Assumption 3' is relaxed to $E(u_t) = 0$ and $E(u_t\mathbf{x}_t) = \mathbf{0}$.
- The assumption of *weak dependence* essentially requires that the random variables become approximately independently distributed or uncorrelated when the amount of time separating them becomes large. This assumption ensures that some versions of the LLN and CLT can be applied to sample averages.

- Consistency of OLS does not require strict exogeneity of the explanatory variables.

Example 1.1. Static model:

$$y_t = \beta_0 + \beta_1 z_{1t} + \beta_2 z_{2t} + u_t.$$

- We can set $\mathbf{x}_t = (z_{1t}, z_{2t})'$. Assumption 3' requires $E(u_t|z_{1t}, z_{2t}) = 0$, which rules out omitted variables that are correlated with any functions of z_{1t} or z_{2t}.
- Assumption 3' does not rule out the correlation between u_{t-1} and z_{1t}. This type of correlation (or feedback effect) could arise if z_{1t} depends on y_{t-1}.
- For instance, z_{1t} may be a policy variable, such as the monthly percentage change in the money supply, and this change may depend on last month's inflation rate (y_{t-1}).
- Nothing guarantees that u_t are uncorrelated over time. If we expect the presence of feedback effect, the errors will usually be serially correlated.

Example 1.2. Finite distributed lag model (FDL or DL):

$$y_t = \alpha + \beta_0 z_t + \beta_1 z_{t-1} + \beta_2 z_{t-2} + u_t. \tag{1}$$

- For example, an advertising campaign may influence sales for some time after it ends, with monthly sales depending on advertising expenditures from the previous few months.
- A natural assumption is that the expected value of u_t given *current and all past* values of z is zero:

$$E(u_t|z_t, z_{t-1}, z_{t-2}, z_{t-3}, \ldots) = 0, \tag{2}$$

which, together with (1), implies that

$$E(y_t|z_t, z_{t-1}, z_{t-2}, z_{t-3}, \ldots) = E(y_t|z_t, z_{t-1}, z_{t-2}) = \alpha + \beta_0 z_t + \beta_1 z_{t-1} + \beta_2 z_{t-2}.$$

That is, no further lags of z affect the expected value of y_t after z_t, z_{t-1}, and z_{t-2} have been included. Assumption 3' clearly satisfies equation (2).

- As in static models, Assumption 3' does not rule out feedback from y to future values of z, and u can be serially correlated.

- Model (1) can be generalized to

$$y_t = \alpha + \beta_0 z_t + \beta_1 z_{t-1} + \beta_2 z_{t-2} + \ldots + \beta_q z_{t-q} + u_t.$$

  The effect of a unit change in z on y after h periods is given by β_h, which is called the h-period *dynamic multiplier*. The cumulative effect of a unit change in z on y over the next h periods is given by $\sum_{j=0}^{h} \beta_j$, which is called the h-period *cumulative dynamic multiplier*. The sum of all individual dynamic multipliers, $\sum_{j=0}^{q} \beta_j$, is called the *long-run cumulative dynamic multiplier*.

Example 1.3. Autoregressive model (AR):

$$y_t = \phi_0 + \phi_1 y_{t-1} + u_t.$$

- Under the assumption

$$E\left(u_t | y_{t-1}, y_{t-2}, \ldots\right) = 0, \tag{3}$$

  we have

$$E\left(y_t | y_{t-1}, y_{t-2}, \ldots\right) = E\left(y_t | y_{t-1}\right) = \phi_0 + \phi_1 y_{t-1},$$

  which means that once y_{t-1} has been controlled for, no further lags of y affect the expected value of y_t. Assumption 3' holds under (3).

- The strict exogeneity assumption needed for the unbiasedness of OLS cannot hold, because u_t is correlated with y_t.

- Assumption TS 5' must hold under (3). To see this, note that $\mathbf{x}_t = y_{t-1}$ in this model. We need to show that $E\left(u_t u_s | y_{t-1}, y_{s-1}\right) = 0$. Without loss of generality, suppose $s < t$, and notice that u_s is a function of y_s, which occurs before time t. Then by (3) and the

law of iterated expectations, we have $E(u_t|u_s, y_{t-1}, y_{s-1}) = 0$, which implies

$$E(u_t u_s|u_s, y_{t-1}, y_{s-1}) = u_s E(u_t|u_s, y_{t-1}, y_{s-1}) = 0,$$

and then $E(u_t u_s|y_{t-1}, y_{s-1}) = 0$ readily follows.

- If we replace (3) with the assumption $E(u_t|y_{t-1}) = 0$ to begin with, then the OLS is still consistent, but the errors can be serially correlated because

$$E(u_t u_{t-1}) = E[u_t(y_{t-1} - \phi_0 - \phi_1 y_{t-2})] = \phi_1 E(u_t y_{t-2}),$$

which is not necessarily zero.

Example 1.4. Autoregressive distributed lag model (ARDL or ADL(p, q)):

$$y_t = \phi_0 + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \ldots + \phi_p y_{t-p} + \beta_0 z_t + \beta_1 z_{t-1} + \ldots + \beta_q z_{t-q} + u_t,$$

where

$$E(u_t|y_{t-1}, y_{t-2}, \ldots, z_t, z_{t-1}, z_{t-2}, \ldots) = 0. \tag{4}$$

- Assumption (4) implies that no further lags of either y or z belong to the model.
- The model is said to be stable if all the roots of the characteristic equation $1 - \phi_1 z - \phi_2 z^2 - \ldots - \phi_p z^p = 0$ lie outside the unit circle (i.e., $|z| > 1$).
- The model can be extended to allow for multiple predictors, $z_1, z_2, \ldots, z_k$, with possibly different numbers of lags.

2 Serial Correlation in Time Series Regressions

2.1 Properties of OLS with Serially Correlated Errors

- OLS is still unbiased (consistent) if Assumptions 1–3 (Assumptions 1'–3') hold regardless of serial correlation.

- R-squared and adjusted R-squared are still valid if data are stationary and weakly dependent ($Corr\left(y_t, y_{t+h}\right)$ goes to zero sufficiently fast as $h \to \infty$).Trends and seasonality in the mean of the dependent variable should be accounted for in computing an R-squared.
- The usual OLS standard errors and test statistics are invalid if there is serial correlation.
- OLS is no longer efficient in the presence of serial correlation.
- To illustrate the last two points, consider the simple regression model

$$y_t = \beta_0 + \beta_1 x_t + u_t, \quad \text{for } t = 1, 2, \ldots, T, \tag{5}$$

with *AR(1) serial correlation* for the error terms:

$$u_t = \rho u_{t-1} + \varepsilon_t, \quad |\rho| < 1,$$

where $\varepsilon_t \sim WN\left(0, \sigma_\varepsilon^2\right)$. Suppose that the other Gauss-Markov assumptions hold, and, to simplify the formula, assume that the sample average of x_t is zero. Then the OLS estimator of β_1 can be written as

$$\hat{\beta}_1 = \beta_1 + \mathrm{SST}_x^{-1} \sum_{t=1}^{T} x_t u_t,$$

where $\mathrm{SST}_x = \sum_{t=1}^{T} x_t^2$, and it follows that the variance of $\hat{\beta}_1$ (conditional on $\mathbf{X}$) is given by

$$\begin{aligned} Var\left(\hat{\beta}_1\right) &= \mathrm{SST}_x^{-2} Var\left(\sum_{t=1}^{T} x_t u_t\right) \\ &= \mathrm{SST}_x^{-2} \left[\sum_{t=1}^{T} Var\left(x_t u_t\right) + 2 \sum_{t-1}^{T-1} \sum_{h=1}^{T-t} E\left(x_t x_{t+h} u_t u_{t+h}\right)\right] \qquad (6) \\ &= \sigma^2 / \mathrm{SST}_x + 2\left(\sigma^2 / \mathrm{SST}_x^2\right) \sum_{t-1}^{T-1} \sum_{h=1}^{T-t} \rho^h x_t x_{t+h}, \qquad (7) \end{aligned}$$

where $\sigma^2 = Var\left(u_t\right)$, and we have used the fact that $E\left(u_t u_{t+h}\right) = Cov\left(u_t, u_{t+h}\right) = \rho^h \sigma^2$. Note that the first term in (7) is the familiar OLS variance formula under the Gauss-Markov

assumptions.

- This example shows that the usual variance estimator will generally be biased when $\rho \neq 0$. In most economic applications, $\rho > 0$, and the independent variables are often positively correlated over time. Therefore, the usual OLS variance formula σ^2/SST_x often understates the true variance. The usual statistics for hypothesis testing, such as t, F, and LM statistics, are also invalid.

- Is OLS necessarily inconsistent in the presence of lagged dependent variables and serially correlated errors? The answer is no. OLS can still be consistent. For example,

$$y_t = \phi_0 + \phi_1 y_{t-1} + u_t,$$
$$E\left(u_t | y_{t-1}\right) = 0,$$

  which satisfies Assumption 3'. Without further assumptions, $\{u_t\}$ can be serially correlated.

- However, the OLS estimators of an AR(1) model with AR(1) serial correlation are inconsistent. To see this, consider

$$y_t = \phi_0 + \phi_1 y_{t-1} + u_t, \tag{8}$$

  where the error terms are assumed to satisfy

$$u_t = \rho u_{t-1} + \varepsilon_t, \quad |\rho| < 1, \tag{9}$$

$$E\left(\varepsilon_t | u_{t-1}, u_{t-2}, \ldots\right) = E\left(\varepsilon_t | y_{t-1}, y_{t-2}, \ldots\right) = 0.$$

  Note that $Cov\left(y_{t-1}, u_t\right) = \rho Cov\left(y_{t-1}, u_{t-1}\right) + Cov\left(y_{t-1}, \varepsilon_t\right) = \rho Cov\left(y_{t-1}, u_{t-1}\right)$, which is not zero unless $\rho = 0$. Therefore, OLS estimators of ϕ_0 and ϕ_1 are inconsistent.

- *Exercise*: Show that model (8)–(9) implies that y_t actually follows an AR(2) process. That is, the expected value of y_t, given all past y, depends on two lags of y.

- Serial correlation in a dynamic model indicates that the dynamics have not been completely specified.

2.2 Serial Correlation-Robust Inference After OLS

- Return to model (5) but now allow for arbitrary serial correlation. For convenience, let $v_t = x_t u_t$ and assume $\{v_t\}$ is stationary. Using the result in (6), we have

$$\begin{aligned} Var\left(\hat{\beta}_1\right) &= \left(\frac{1}{SST_x/T}\right)^2 \left[TVar\left(v_t\right) + 2\left(T-1\right)Cov\left(v_t, v_{t-1}\right) + 2\left(T-2\right)Cov\left(v_t, v_{t-2}\right)\right. \\ &\quad \left. + \ldots + 2Cov\left(v_t, v_{t-T+1}\right)\right]/T^2 \\ &= \left[\frac{1}{T}\frac{\sigma_v^2}{\left(\sigma_x^2\right)^2}\right] f_T, \end{aligned} \tag{10}$$

where $\sigma_x^2 = Var\left(x_t\right) = SST_x/T$, $\sigma_v^2 = Var\left(v_t\right)$, and

$$f_T = 1 + 2\sum_{h=1}^{T-1}\left(\frac{T-h}{T}\right)\rho_h,$$

with $\rho_h = Corr\left(v_t, v_{t-h}\right)$.

- The term in the square bracket in (10) is the standard formula for the heteroskedasticity-robust variance of $\hat{\beta}_1$ in the absence of serial correlation. Let $\hat{\sigma}^2_{\hat{\beta}_1}$ denote its usual estimator. Then the serial correlation-robust variance estimator of $\hat{\beta}_1$ is $\tilde{\sigma}^2_{\hat{\beta}_1} = \hat{\sigma}^2_{\hat{\beta}_1}\hat{f}_T$, where

$$\hat{f}_T = 1 + 2\sum_{h=1}^{m}\left(1 - \frac{h}{m+1}\right)\hat{\rho}_h,$$

with $\hat{\rho}_h = \sum_{t=h+1}^{T}\hat{v}_t\hat{v}_{t-h}/\left(\sum_{t=1}^{T}\hat{v}_t^2\right)$, $\hat{v}_t = x_t\hat{u}_t$, and $\hat{u}_t = y_t - \hat{\beta}_0 - \hat{\beta}_1 x_t$. The integer m is called the *truncation parameter* or *truncation lag*.

- Since $\tilde{\sigma}^2_{\hat{\beta}_1}$ is also robust to arbitrary heteroskedasticity, $\tilde{\sigma}_{\hat{\beta}_1} = \sqrt{\tilde{\sigma}^2_{\hat{\beta}_1}}$ is often called **heteroskedasticity and autocorrelation consistent (HAC) standard errors** or **Newey-West standard errors** (Newey and West, 1987).

- For $\hat{f}_T$ to be consistent, the value of m must be sufficiently large, but still much smaller

than T. Stock and Watson (2020) recommend taking m to be the integer part of $0.75T^{1/3}$ in practice. Some software uses the integer part of $4\,(T/100)^{2/9}$ by default, while others have suggested the integer part of $T^{1/4}$.

- The above argument readily extends to x_t with a nonzero mean and multiple regressors.
- R implementation: `NeweyWest()` contained in the `sandwich` package

References:

Wooldridge (2019, Introductory Econometrics), Chs. 10–12

Stock and Watson (2020, Introduction to Econometrics), Ch. 16